



N-Acetylcysteine Intravenous Infusion Protocol for Paracetamol Poisoning

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1. Purpose

This guideline is for Nursing and Medical staff within the Emergency Department (ED) to ensure an appropriate N-Acetylcysteine Intravenous Infusion Protocol is used in a timely manner for paracetamol overdose. Paracetamol overdose can cause severe liver injury and potentially death from fulminant liver failure if it remains untreated.

2. Guidelines

This guideline is relevant for adult and paediatric patients with a single oral ingestion of immediate-release paracetamol $\geq 10\text{g}$ or $\geq 200\text{ mg/kg}$ (whichever is less) and $< 500\text{mg/kg}$, presenting to hospital within 8 hours, where a timed paracetamol level indicates risk of hepatotoxicity. This guideline **does not cover** the management of paracetamol poisoning – expert advice should be sought for this:

[Summary of updated guidelines for management of paracetamol poisoning in Australia and New Zealand \(MJA 2019\)](#)

2.1 Toxicology Advice

It is recommended to seek toxicological advice (**Poisons Information Centre 13 11 26**) in the following situations as modifications to standard management may be required:

- If the history of ingestion or required management is not clear
- Patients presenting with staggered ingestions
- Repeated supratherapeutic ingestions
- Modified-release formulation ingestion
- Massive ingestion $\geq 30\text{g}$ or $\geq 500\text{ mg/kg}$ (whichever is less)
- Paracetamol level is $\geq$ twice the nomogram line
- When presentation is delayed > 8 hours
- Mixed ingestions of more than one agent
- Iatrogenic overdose of IV paracetamol
- Neonatal paracetamol poisonings
- Clinical or laboratory evidence of hepatotoxicity e.g. ALT $> 1000\text{ U/L}$

2.2 Administration of N-Acetylcysteine Intravenous Infusion

N-Acetylcysteine (NAC) is packaged for intravenous infusion in 2g/10ml ampoules (20% solution i.e. 200mg NAC per 1 ml)

NAC is typically infused in a 2-step intravenous infusion (total dose 300mg/kg over 20 hours)

- Initial infusion (Adults):
 - 200mg/kg NAC in 500mls of glucose 5% given over 4 hours
- Second infusion (Adults):
 - 100mg/kg NAC in 1000mls of glucose 5% given over 16 hours

2.3 Dosing Calculator

Obtain weight of patient (kg) to determine dosage

*Adult dosing should be based on **actual body weight rounded up to the nearest 10 kg** with a ceiling weight of **110 kg**

Paediatric dosing based on actual body weight, but with an appropriately reduced [volume of infusion](#)

Calculation or transcription errors may lead to potentially fatal dosing errors. It is recommended that dosing tables (see below) providing the required dose and volume of 20% acetylcysteine by weight, are used to chart each infusion. This precludes the need for calculations and decreases the potential for error

The corresponding volume is removed from the infusion fluid then NAC added to bag. Invert all prepared solutions **at least 10 times** prior to infusing to ensure adequate mixing

[FSHFG Guideline - Acetylcysteine intravenous infusion protocol for paracetamol poisoning](#)
[PCH Medication Management Manual - Acetylcysteine Monograph - Paediatric](#)

	Infusion 1 200mg/kg NAC Added to 500ml 5% Dextrose Given over 4 hours		Infusion 2 100mg/kg NAC Added to 1000ml 5% Dextrose Given over 16 hours	
Patient's weight (kg)*	Dose of NAC (grams)	Volume of NAC (ml of 200mg/ml NAC solution)	Dose of NAC (grams)	Volume of NAC (ml of 200mg/ml NAC solution)
40 kg	8g	40 ml	4g	20 ml
50 kg	10g	50 ml	5g	25 ml
60 kg	12g	60 ml	6g	30 ml
70 kg	14g	70 ml	7g	35 ml
80 kg	16g	80 ml	8g	40 ml
90 kg	18g	90 ml	9g	45 ml
100 kg	20g	100 ml	10g	50 ml
110 kg	22g	110 ml	11g	55ml

2.4 Paediatric Fluid Volumes

The volume and choice of fluid for infusion should be appropriate to age, weight and clinical circumstances of the child. Adolescents usually follow the standard adult protocol as above. NAC is compatible with 0.9 % saline, 0.9% saline + 5% glucose, and 0.45 % saline + 5% glucose

Children or adolescents >50 kg (*recommended ceiling weight 110 kg*):

Infusion stage	NAC dose	Dilute to	Infusion duration
First	200 mg/kg	TOTAL volume 500 mL	4 hours
Second	100 mg/kg	TOTAL volume 1000 mL	16 hours

For children 20 – 50 kg body weight:

Infusion stage	NAC dose	Dilute to	Rate and duration
First	200 mg/kg	TOTAL volume 250 mL	4 hours
Second	100 mg/kg	TOTAL volume 500 mL	16 hours

NOTE: this results in a total of 750 mL of fluid which is inappropriate for smaller children.

For children ≤20 kg body weight:

Infusion stage	NAC dose	Dilute to	Rate and duration
First	200 mg/kg	TOTAL volume 100 mL	4 hours
Second	100 mg/kg	TOTAL volume 250 mL	16 hours

Smaller volume for dilution may be used in younger infants or fluid restricted patients (7mL/kg for the first infusion, 14mL/kg for the second infusion) – seek Toxicology/Paediatric advice

2.5 Contraindications for Procedure

NAC is not indicated for non-steroidal anti-inflammatory drugs (NSAID) overdose or for paracetamol overdose where a timed paracetamol level indicates the patient is low risk for hepatotoxicity

Previous anaphylactoid reaction to NAC is NOT a contraindication for future use

2.6 Potential Adverse Reactions

- **Nausea and vomiting**
 - Common: may be due to either NAC or the paracetamol ingestion itself
 - Management: treat with standard antiemetic therapy
- **Non-allergic Anaphylactoid Reaction (NAAR)**
 - Occur in ~5% of patients: usually during initial infusion, due to non-IgE mediated histamine release (much less common occurrence compared to previous 3-bag NAC regime)
 - May develop flushing, urticarial rash, mild wheeze or hypotension
 - Management: supportive
 - Stop infusion temporarily and seek medical review
 - Consider non-sedating antihistamines
 - Consider bronchodilators for wheeze

- Once reaction settles, infusion can be recommenced at the usual rate

- **Severe life-threatening reactions (Anaphylaxis)**

- Very rare (0.5%)
- May develop bronchospasm, angioedema, dyspnoea, hypotension, shock
- Management: call for help, treat as per Anaphylaxis (i.e. with adrenaline if required)
- Adverse reactions more likely to occur in predisposed individuals, such as patients with asthma
- Once adverse reaction subsides NAC should be recommenced

2.7 Equipment Required

- Standard infusion pump and IV giving set
- Resuscitation equipment accessible **at all times**

2.8 Patient Monitoring

Prior to administration provide patients with education regarding the [adverse effects](#) that they may experience during infusion

Observation of vital signs should be performed before giving infusion and every 15 minutes for the first hour. Increase frequency of observations as indicated by patient's clinical condition

Monitoring with pulse oximetry for first 2 hours of infusion is recommended

2.9 Additional Information

NAC may be given mixed in sodium chloride 0.9% if there are concerns about hyponatraemia or traumatic brain injury

Once an acetylcysteine vial has been punctured the contents may turn from colourless to slightly pink or purple. This does not affect drug potency

By consensus, there are frequent situations where Australian clinical toxicologists will alter the NAC regime by prolonging the infusion or increasing the concentration of the 16-hour infusion of NAC

[Full guidelines for management of paracetamol poisoning in Australia and New Zealand \(MJA 2019\)](#)

3. Relevant standards

NSQHS Standards:

- Clinical Governance
- Medication Safety
- Comprehensive Care
- Recognising and Responding to Acute Deterioration

4. Bibliography

1. [FSHFG Guideline - Acetylcysteine intravenous infusion protocol for paracetamol poisoning](#)
2. [FSH Pharmacy Department – Intravenous Drug Administration Guideline - Acetylcysteine](#)
3. [PCH Medication Management Manual - Acetylcysteine Monograph - Paediatric](#)
4. Chiew AI, Reith D, Pomerleau A, et al. Updated guidelines for the management of paracetamol poisoning in Australia and New Zealand. Med J Aust 2019; doi: 10.5694/mja2.50428; available at https://www.mja.com.au/system/files/issues/212_04/mja250428.pdf, accessed 02/10/2025
5. McNulty R, Lim JME, Chandru P, Gunja N. Fewer adverse effects with a modified two-bag acetylcysteine protocol in paracetamol overdose. Clin Tox 2018; available at <https://www.tandfonline.com/doi/full/10.1080/15563650.2017.1408812>, accessed 02/10/2025

5. Document owner

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6. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	17 10 2025	17 10 2028	Minor amendments only to ensure guideline clearer for use

7. Authorisation

This guideline has been authorised and issued by:

Authorised by	Dr Tania Strickland – Director Clinical Services
Authorisation date	16 10 2025
Published date	17 10 2025